ads1-hw-examples (/github/BenLangmead/ads1-hw-examples/tree/master)
/ hw1_naive_with_rc.ipynb (/github/BenLangmead/ads1-hw-examples/tree/master/hw1_naive_with_rc.ipynb)

In [1]:
```
# Implement the naive_with_rc function
from naive_with_rc import naive_with_rc
```

Example 1

In [2]:
```
p = 'CCC'
ten_as = 'AAAAAAAAAA'
t = ten_as + 'CCC' + ten_as + 'GGG' + ten_as
occurrences = naive_with_rc(p, t)
print(occurrences)
```

```
[10, 23]
```

Example 2

In [3]:
```
p = 'CGCG'
t = ten_as + 'CGCG' + ten_as + 'CGCG' + ten_as
occurrences = naive_with_rc(p, t)
print(occurrences)
```

```
[10, 24]
```

Example 3

In [4]:
```
# Phi-X genome
!wget http://d396qusza40orc.cloudfront.net/ads1/data/phix.fa
```

```
--2015-08-24 21:11:45--  http://d396qusza40orc.cloudfront.net/ads1/data/phix.fa
Resolving d396qusza40orc.cloudfront.net... 204.246.169.113, 54.230.39.39, 54.230.37.173,
...
Connecting to d396qusza40orc.cloudfront.net|204.246.169.113|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 5528 (5.4K) [application/octet-stream]
Saving to: 'phix.fa'

phix.fa             100%[====================>]   5.40K  --.-KB/s   in 0s

2015-08-24 21:11:45 (13.1 MB/s) - 'phix.fa' saved [5528/5528]
```

In [5]:
```
def readGenome(filename):
    genome = ''
    with open(filename, 'r') as f:
        for line in f:
            # ignore header line with genome information
            if not line[0] == '>':
                genome += line.rstrip()
    return genome
```

In [6]:
```
phix_genome = readGenome('phix.fa')
```

In [7]:
```
occurrences = naive_with_rc('ATTA', phix_genome)
```

In [8]:
```
print('offset of leftmost occurrence: %d' % min(occurrences))
```

```
offset of leftmost occurrence: 62
```

In [9]:
```
print('# occurrences: %d' % len(occurrences))
```

```
# occurrences: 60
```